3.1 → Risk classification (cont...)

→ Univariate classification

→ Goal → Determine relativities for each level of a rating variable

⇒ ie) determines rate differentials by analyzing historical experience of each level of a rating variable independently

→ There are three approaches

→ Pure premium approach →

(1) Territory	(2) Exposures	(3) Loss & ALAE	(4) Indicated Pure Premium	(5) Indicated Relativity	(6) Indicated Relativity to Base
1	300	$15,698.08	$52.33	0.7696	0.7231
2	390	$28,221.07	$72.36	1.0643	1.0000
3	310	$24,072.96	$77.65	1.1421	1.0731
Total	1,000	$67,992.11	$67.99	1.0000	0.9396

(5) = using $(4_i)/(4\ total)$

(6) → $(5_i)/(5_2)$ or $(4_i)/(4_2)$

→ LR approach →

(1) Territory	(2) Premium at Current Rate Level	(3) Loss & ALAE	(4) Loss & ALAE Ratio	(5) Indicated Change Factor	(6) Current Relativity	(7) Indicated Relativity	(8) Indicated Relativity to Base
1	$25,740.66	$15,698.08	60.99%	1.0763	0.6000	0.6458	0.6354
2	$49,002.80	$28,221.07	57.59%	1.0164	1.0000	1.0164	1.0000
3	$45,256.64	$24,072.96	53.19%	0.9388	1.3000	1.2204	1.2007
Total	$120,000.10	$67,992.11	56.66%	1.0000			

(2) necessary

(5) = $(4_i)/(4\ total)$

(7) = (5) * (6) → factor in-place now

(8) = $(7_i)/(7_2)$

→ Adjusted PP approach → adjusts for any distributional bias, like the LR method

(0) factor	(1) Adj exposures	(2) loss + LAE	(3) Indicated adj PP	(5) Indicated relativity	(5) Indicated relativity to base
factor of interest	sumproduct of current factor over exp by other factor	xLAE	(2)/(1)	(3)/(3 total)	(4)/(4 base)

→ Changing class relativities

→ Ind class rate change = Ind avg rate change factor * Rel change factor * OBF − 1

Ind avg rate change factor → =1 if revenue neutral

Rel change factor → cred-wtd + rebased

OBF → accounts for the change in the average relativity that we unintentionally cause by creating new class relativities for each class.

→ wtd avg rel = $\frac{\text{Sumproduct (current factor, adj exp)}}{\text{Sum (adj exp)}}$

⇒ OBF = $\frac{\text{wtd avg rel current}}{\text{wtd avg rel proposed}}$

→ Univariate classification w/ credibility

→ GOAL → using credibility enhances the reliability of estimates derived from univariate analysis

→ Blending class-specific data w/ broader, more stable datasets mitigates the impact of random variations + ensures more accurate rate indications

→ credibility estimate = Credibility * Experience + (1 − Credibility) * complement

need same base

→ to rebase a set of relativities to the total:

→ 1) Calculate the weighted avg of the relativities

→ 2) Divide each relativity by the avg calculated in (1)

→ 3) the re-based relativities should have a weighted-avg = 1

→ Pure premium + adjusted pure premium approaches →

(1) Territory	(2) Exposures	(3) Current Relativity	(4) Normalized Current Relativity	(5) Indicated Relativity to Total	(6) Credibility	(7) Cred-Wtd Indicated Relativity	(8) Cred-Wtd Indicated Relativity to Base
1	300	0.600	0.6166	0.7696	0.7746	0.7351	0.6935
2	390	1.000	1.0277	1.0643	0.8832	1.0600	1.0000
3	310	1.300	1.3361	1.1421	0.7874	1.1833	1.1164
Total	1,000	0.9730	1.0000	1.0000			

(6) calculated w/ a given formula

(note for these examples, the complement of credibility is no change)

- (3)total = [(3) × (2)] ÷ (2)total
- (4) = (3) ÷ (3)total
- (7) = (6) × (5) + [1 - (6)] × (4)
- (8) = (7) ÷ (7)base

if adjusted PP method, use adjusted exposures instead ⇒ rest is same

→ LR approach →

(1) Territory	(2) Current Relativity	(3) Indicated Change Factor	(4) Credibility	(5) Cred-Wtd Indicated Change Factor	(6) Cred-Wtd Indicated Relativity	(7) Cred-Wtd Indicated Relativity to Base
1	0.600	1.0763	0.7746	1.0591	0.6355	0.6264
2	1.000	1.0164	0.8832	1.0145	1.0145	1.0000
3	1.300	0.9388	0.7874	0.9518	1.2373	1.2197

(instead of ind relativity)

- (5) = (4) × (3) + [1 - (4)] × 1.0000
- (6) = (5) × (2)
- (7) = (6) ÷ (6)base

difference ??

→ If we were to rebase the relativities in the LR approach, here is what it would look like:

(1) Territory	(2) Premium at Current Rate Level	(3) Premium at CRL at Base	(4) Current Relativity	(5) Normalized Current Relativity	(6) Indicated Relativity	(7) Normalized Indicated Relativity	(8) Credibility	(9) Cred-Wtd Indicated Relativity	(10) Cred-Wtd Indicated Relativity to Base
1	$25,740.66	$42,901.10	0.600	0.6336	0.6458	0.6820	0.7746	0.6710	0.6264
2	$49,002.80	$49,002.80	1.000	1.0560	1.0164	1.0733	0.8832	1.0713	1.0000
3	$45,256.64	$34,812.80	1.300	1.3728	1.2204	1.2887	0.7874	1.3066	1.2197
Total	$120,000.10	$126,716.70	0.9470	1.0000	0.9470	1.0000			

- (3) = (2) ÷ (4)
- (4)total = [(4) × (3)] ÷ (3)total
- (5) = (4) ÷ (4)total
- (6)total = [(6) × (3)] ÷ (3)total
- (7) = (6) ÷ (6)total
- (9) = (8) × (7) + [1 - (8)] × (5)
- (10) = (9) ÷ (9)base

→ Notes → credibility is not based on adj exposures when calculating relativities

→ have to normalize current factors before using them in a credibility calculation

⇒ normalized rel = $\frac{\text{current rel}}{\text{wtd avg rel}}$ (formula for this is written above)

3.2 → Multivariate classification

→ Minimum bias procedures → iteratively standardized univariate approaches that account for an uneven mix of business. Each procedure involves the following steps:

→ 1) Select a rating structure, such as additive, multiplicative, or combined

→ 2) Choose a bias function, which measures the difference between the procedure's observed loss statistics + indicated loss statistics. Examples of bias functions are balance principle, least squares, χ^2, + maximum likelihood functions.

→ 3) Weighting both sides of the bias function by the exposures in each cell to adjust for uneven mix of business.

→ 4) "minimizing the bias", which is accomplished by equating the sum of indicated pure premium + the sum of the weighted observed loss costs for every level of every rating variable.

→ Sequential analysis → Pick first variable + find relativities to base like usual

⇒ Perform adj PP approach for next variable using results from above

⇒ repeat w/ new variable + new results

adjusts based on exposure distribution

⇒ If deciding which variables to adjust for in a univariate classification problem, think logically ⇒ don't have to "cross" every variable ⇒ use context + state decision

→ Comparison

→ In contrast to minimum bias procedures, which iterate until convergence is achieved, sequential analysis is not iterative.

→ The main drawbacks of sequential analysis is that it does not have a closed-form solution, resulting in varying outcomes based on the order of rating variables in the sequence.

→ Multivariate classification

→ Why needed?

→ univariate methods

→ offer the advantage of transparency, but this benefit is outweighed by the inaccuracies of the method.

→ Results of univariate methods are not only distorted by distributional bias, but also by unsystematic effects (noise).

→ Commonly, univariate analysis lacks information about the certainty of results

→ Minimum bias procedures

→ Are an improvement over univariate approaches as they account for an uneven mix of business while maintaining transparency

→ Do not include diagnostics

→ Benefits of multivariate methods

→ 1) Take into account all rating variables simultaneously + automatically adjust for exposure correlations between rating variables

→ 2) Attempt to capture only systematic effects (signals) + ignore unsystematic effects (noise).

→ 3) Generate model diagnostics, offering additional information about the certainty of results + appropriateness of the fitted model

→ 4) Allow consideration of the interaction between two or more rating variables, known as response correlation

→ GLMs → Additional benefits

→ Transparent

→ model output includes parameter estimates for every level of each explanatory variable w/in the model, along w/ a range of statistical diagnostics

→ Data type used in analysis → typically performed on loss cost data, preferably frequency + severity data separately, instead of loss ratio data b/c:

⇒ (one reason) no standard distribution for modelling loss ratios

→ Uses of Diagnostics

→ 1) To determine if a predictive variable has a systematic effect on insurance losses + should thus be retained in the model.

→ 2) To evaluate the modeler's assumptions regarding the link function + the error term.

→ Common diagnostics → 1) standard errors

→ 2) measures of deviance

→ 3) Practical diagnosis

→ Model validating

→ other than running statistical diagnostics, an analyst should also evaluate how well the model validates (e.g. run on holdout dataset)

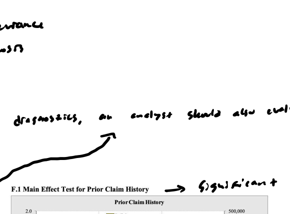
→ Tests → 1) Main effects test

→ 2) Consistency test

→ 3) Statistical test

→ 4) Judgemental

→ 5)? validation tests



→ significant

→ significant still

★ → Final decision → All four tests suggest the rating variable is predictive + should be included in the model (+ ultimately the rating algorithm)

3.3 → Special classification

→ Increased limits ratemaking → the process of determining indicated increased limit factors (ILFs)

→ why needed? → limited data leads to volatile results. multivariate approaches are better for sparse data, but can produce illogical rating factors (e.g. decrease in factor for a higher limit)

⇒ have specialized approaches

→ Assumptions → 1) All underwriting expenses are variable

→ 2) The variable expense + profit provisions do not vary by limit

→ 3) Frequency + severity are independent

→ 4) Frequency is the same for all limits

→ Standard ILF approach

→ $Rate_H = ILF_{(H)} \times Rate_{Base}$

$ILF_{(H)} = \frac{LAS(H)}{LAS(B)}$ (B = Base)

→ Censored losses

	Limit: Base = H_1 claims	losses	H_2 claims	losses	H_3 claims	losses
$0 < x \le H_1$						
$H_1 < x \le H_2$						
$H_2 < x$						

→ $LAS(H_1) = \frac{\sum_{H_i, i=1,2,3}\left[\sum_{0<x\le H_1} losses\right] + \sum_{H_i, i\in 2,3} H_1 \times (\#\ claims\ H_1 \le x \le H_3)}{\sum_{All} claims}$

< row in that layer > + < capped above layers >

↓ severity assuming all losses are capped at H_1

$LAS(H_2) > LAS(H_1)$ ⇐ but doesn't have to be $> H_1$

→ $LAS((H_2 - H_1)\ xs\ H_1) = \frac{\left(\sum_{H_i, i=2,3}\left[\sum_{H_1<x\le H_2} losses\right] + \sum_{H_3} H_2 \times \#\ claims\ (H_2 \le x < H_3)\right) - H_1 \times \#\ claims\ used\ in\ (\ )}{\sum claims\ used\ in\ (\ )}$

< row in that layer > + < capped above layers > − < H_1 accounted for >

→ $LAS(H_2) = LAS(H_1) + P(H_1 \le x < H_2) \times LAS((H_2 - H_1)\ xs\ H_1)$

$P(H_1 \le x < H_2) = \frac{\sum claims\ used\ in\ (\ )}{\sum_{H_i, i=2,3} claims}$

→ ILFs are always relative to the base limit!

⇒ $ILF(H_2) = \frac{LAS(H_2)}{LAS(H_1)}$

→ $LAS((H_3 - H_2)\ xs\ H_2) = \frac{\sum_{H_3}\left[\sum_{H_2<x\le H_3} losses\right] - H_2 \times \#\ claims\ used\ in\ (\ )}{\#\ claims\ used\ in\ (\ )}$

< raw > − < H_2 accounted for >

→ $LAS(H_3) = LAS(H_2) + P(H_2 \le x < H_3) \times LAS((H_3 - H_2)\ xs\ H_2)$

$P(H_2 \le x < H_3) = \frac{\#\ claims\ used\ in\ (\ )}{\sum_{H_3} claims}$

⇒ $ILF(H_3) = \frac{LAS(H_3)}{LAS(H_1)}$

→ Increased limits excess ratemaking = losses capped at lower limit (using only policies at or above that limit) $H_i \le x \le H_{i+1}$ * $\frac{ILF(H_{i+1}) - ILF(H_i)}{ILF(H_i)}$ (complement of cred)

⇒ Accurate? not if difference in loss severity dist

unbiased? Biased if — — —

independent? generally

Available? Practical if data is available

Easy to compute? yes

logical? maybe

→ Deductible pricing (relativities found via the loss elimination ratio (LER) approach

→ Assuming all expenses are variable + the variable expenses + profit are a constant percentage of premium, the indicated deductible relativity for going from one deductible level, B, to a higher one, D can be calculated as

→ Indicated deductible relativity = $\frac{(L+E_L)_D}{(L+E_L)_B}$ = <...> = 1 − LER(D)

where $(L+E_L)_B$ + $(L+E_L)_D$ are the losses + LAE net of the deductibles B + D, respectively.

→ treat like a shock loss problem, where deductible is the threshold (it's just gonna be a low H)

* IF B=0 ⇒ full coverage

→ The loss elimination ratio → $LER(D) = \frac{(L+E_L)_B - (L+E_L)_D}{(L+E_L)_B}$ = $\frac{\text{losses + LAE eliminated by deductible D}}{\text{total ground-up losses + LAE}}$

→ represents the proportion of losses eliminated by going from one deductible to another. Speak differently, the loss elimination ratio is one minus the proportion of losses that remain after switching deductible.

→ Two methods to calculate LER:

→ Method 1 → The ratio of losses eliminated by the deductible + the total ground-up losses

→ Method 2 → One minus the ratio of the losses that remain after applying the deductible to the ground-up losses

→ Insurance to value

→ Issues → 1) Insured's perspective → The insurance payout will not adequate to cover a total or near-total loss

→ 2) If the insurer bases its rate on the assumption that all homes are insured to their full value, the premium charged will not be sufficient to cover the expected loss on underinsured homes

→ Equitable rates → Coinsurance shares losses w/ underinsureds so ITV insureds do not have excessive rates

→ Adequate rates → Coinsurance reduces the loss by forcing underinsureds to take a portion of the loss in order to reach an adequate rate

→ Coinsurance + rate per $1000 of coverage

→ Calculations → V = true value

→ F = coverage amount, F < V + cV

→ cV = coinsurance requirement

⇒ PP / (coverage / 1000)

capped at coverage value (i.e. if value of home > cov value, losses are capped at cov value)

→ a = apportionment ratio → % of property insured up to

→ I = Amount insurer pays = apportionment ratio * loss

capped at face value

→ e = Coinsurance penalty → Total possible paid by the insurer − Indemnity payment

* remember when a loss is greater than the face value + a home is underinsured, the policyholder is responsible for the leftover damages

3.4 → Credibility

→ Necessary criteria for measures of credibility

→ 1) $0 \le Z \le 1$

→ 2) As n increases, Z should increase

→ 3) As n increases, Z should increase at a decreasing rate

→ Method 1 → Classical credibility

→ Estimate = (Z * observed exp) + (1 − Z) related experience

→ Simplifying assumptions:

→ Exposures are homogeneous

→ claim occurrence follows a Poisson distribution

→ There is no variation in loss size

→ Classical credibility has three main advantages:

→ 1) It is the most commonly used credibility method

→ 2) The data required is readily available

→ 3) The calculations are very straightforward

→ Calculations → Expected # of claims: $E(y) = \left(\frac{z_{(p+1)/2}}{k}\right)^2$

$Z = \sqrt{\frac{Y}{E(Y)}}$, where $Y < E(Y)$

⇒ # exposures = # claims ÷ Freq

Freq = $\frac{\#\ claims}{\#\ exposures}$

→ Method 2 → Bühlmann credibility (generally accepted)

→ calculates the credibility-weighted estimate as follows:

Estimate = Z * observed experience + (1 − Z) * prior mean

→ under Bühlmann credibility, Z is defined as follows:

$Z = \frac{N}{N+K}$

where

→ N = number of observations

→ $K = \frac{\text{the expected value of process variance (EVPV)}}{\text{the variance of hypothetical means (VHM)}}$

→ Method 3 → Bayesian analysis (very complex ⇒ not common)

→ Complements of credibility

→ Desirable qualities of a complement of credibility

→ There are six desirable qualities for a complement of credibility.

→ 1) Accurate → The complement should be close to the expected losses being estimated

→ 2) Unbiased → The complement should not be consistently higher or lower than the observed experience.

→ 3) Statistically independent from the base statistic → Being statistically independent from the base statistic ensures that any errors in the base statistic do not compound.

→ 4) Available → The data required to compute the complement should be readily available for the complement to be practical.

→ 5) Easy to compute → The calculations of the complement should be relatively straightforward, so that justification can be provided when required.

→ 6) Logical relationship + base statistic → The complement should have a logical relationship to the observed experience so that its use can be easily justifiable.

→ First dollar ratemaking

→ There are six commonly used methods for developing complements for first dollar ratemaking:

→ 1) Loss costs of a larger group that includes the group being rated

→ e.g.) State X + CW data (including X)

→ Highlights → Accurate (based on more data), mostly unbiased, Independent (if subject experience is only a small portion)

→ 2) Loss costs of a larger related group

→ e.g.) using losses on cars as complement for losses of trucks

→ Highlights → Maybe inaccurate, Biased, Independent

→ If complement is "All other states" + given CW data, then need to subtract out the state of interest's numbers

→ 3) Rate change from the larger group applied to the present rates

→ Reduces bias from (2)

→ C = current loss cost of subject experience * $\frac{\text{larger group Adjusted loss cost}}{\text{larger group current avg loss cost}}$

C → complement

→ 4) Harwayne's method

→ unbiased b/c it adjusts for distributional differences

→ Highlights → unbiased, computation is complicated, mostly independent

(a) State	(b) Overall PP assuming State A's exposure distribution	(c) Adjustment factor	(d) Class 1 PP	(e) Adjusted class 1 PP	(f) Class 1 exposure
A		$\frac{(B)_A}{(B)}$	(bring in)	(c) * (d)	(bring in)
B	$\frac{\text{sumproduct}(\hat{exp}, PP)}{\text{sum}(\hat{exp})}$	(B → A ⇒ A/(B)) A relative to B		* multiply	
C					

wtd avg ⇒ $\frac{\text{Sumproduct (e), (f)}}{\text{Sum (e)}}$

→ 5) Trended present rates

→ used if no larger group available

→ PP approach → C = Present rate * loss trend factor * $\frac{\text{Prior indicated loss cost}}{\text{loss cost implemented last review}}$

C → complement

→ LR approach → $C = \left(\frac{\text{loss trend factor}}{\text{premium trend factor}}\right)^t \left(\frac{\text{prior indicated rate change factor}}{\text{— implemented — — —}}\right)$

t → trend period length (old indication effective date to new indication eff date)

can leave as factor OR subtract 1

just need to be consistent when doing

cred-weighted indicated rate change factor = Indicated factor (Z) + complement factor (1 − Z)

OR

— — — % = Indicated rate change % (Z) + complement % (1 − Z)

i.e. C − 1 minus 1

→ Highlights → unbiased (since pure trended loss costs are unbiased), accurate, easy to compute

→ 6) Competitor's rates

→ Used in new / small companies

→ Highlights → Inaccurate + unbiased (due to company differences), may be difficult to obtain

→ Mix of business is the main difficulty in comparing w/ competitors' rates (to correct for this, can compare rates for particular segments)